

HOW OLD ARE FOSSILS?

BY

SHARAT K. ROY

ASSISTANT CURATOR OF INVERTEBRATE PALEONTOLOGY



GEOLOGY

LEAFLET 9

FIELD MUSEUM OF NATURAL HISTORY
CHICAGO

1927

HOW OLD ARE FOSSILS?

BY

SHARAT K. ROY

ASSISTANT CURATOR OF INVERTEBRATE PALEONTOLOGY



GEOLOGY

LEAFLET 9

FIELD MUSEUM OF NATURAL HISTORY

CHICAGO

1927

The Project Gutenberg eBook of How old are fossils?

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: How old are fossils?

Author: Sharat Kumar Roy

Release date: November 21, 2023 [eBook #72188]

Language: English

Original publication: Chicago, IL: Field Museum of Natural History, 1927

Other information and formats: www.gutenberg.org/ebooks/72188

Credits: The Online Distributed Proofreading Team at <https://www.pgdp.net>
(This file was produced from images generously made available
by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK HOW OLD ARE
FOSSILS? ***

HOW OLD ARE FOSSILS?

BY
SHARAT K. ROY
ASSISTANT CURATOR OF INVERTEBRATE PALEONTOLOGY



GEOLOGY
LEAFLET 9

FIELD MUSEUM OF NATURAL HISTORY
CHICAGO

1927

**LIST OF GEOLOGICAL LEAFLETS
ISSUED TO DATE**

No. 1.	Model of an Arizona Gold Mine	\$.10
No. 2.	Models of Blast Furnaces for Smelting Iron	.10
No. 3.	Amber—Its Physical Properties and Geological Occurrence	.10
No. 4.	Meteorites	.10
No. 5.	Soils	.10
No. 6.	The Moon	.10
No. 7.	Early Geological History of Chicago	.25
No. 8.	Agate	.50
No. 9.	How old are Fossils?	.25

D. C. DAVIES, DIRECTOR

FIELD MUSEUM OF NATURAL HISTORY
CHICAGO, U.S.A.

LEAFLET 9.

PLATE I.



A VIEW OF THE GRAND CANYON OF THE COLORADO RIVER,
ARIZONA.

ZOROASTER IN BACKGROUND.

Photograph by O. C. Farrington.

FIELD MUSEUM OF NATURAL HISTORY

DEPARTMENT OF GEOLOGY

CHICAGO, 1927

LEAFLET

NUMBER 9

How old are Fossils?

“How old is that fossil and how do you know it?” is a question frequently asked by visitors going through the hall of fossils in the Museum. A precise answer to such a question is impossible and an adequate one demands a longer time than can usually be afforded. The consequences of inadequate explanation have often proved to be unsatisfactory. The visitor becomes skeptical and instead of taking interest in the subject, he seems to be confirmed in his doubts.

In this leaflet is given a condensed, general statement of methods of determining the age of ancient life. The information is drawn from the works of various authors, especially Barrell’s “Rhythms and Measurements of Geologic Time.” It is intended for those who are interested in the age of past life and yet do not intend an exhaustive study of the subject, nor have free and easy access to its literature.

The birth of life was the most momentous occasion in the history of the earth. When one considers the myriads of evidences unearthed by paleontologists and paleobotanists, they seem to leave no room to doubt the great conception that the life of the land has emerged from the sea. It is, therefore, only a natural impulse to look for the remains of this life in the rocks laid down by the ancient seas and to wonder at the vastness of time behind them.

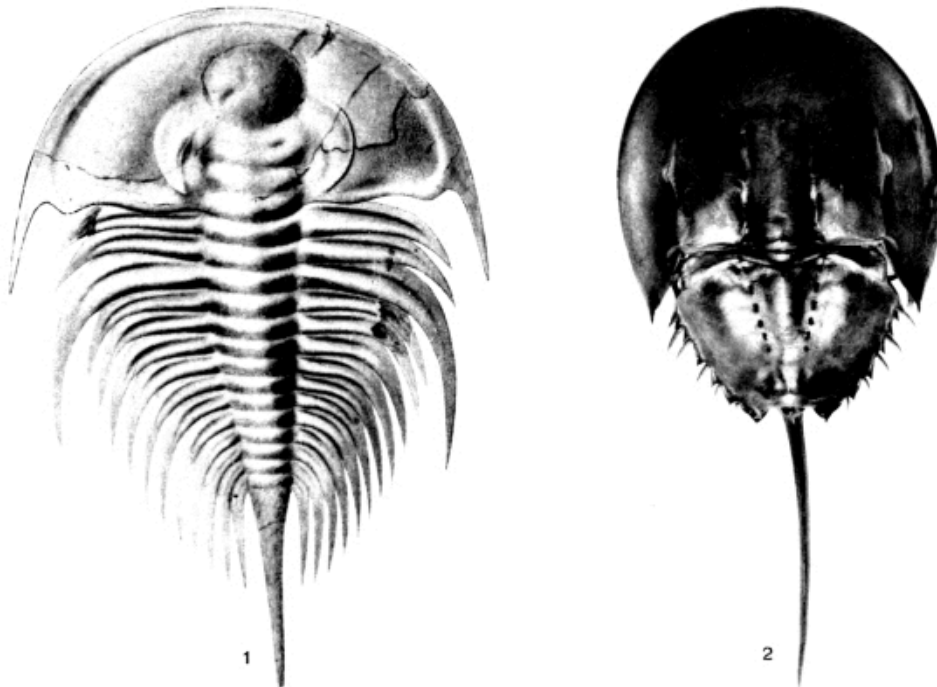
Since traces of the lowest forms of life have been found in practically the oldest known sedimentary strata, the problem of determining the age of life necessarily involves the determination of the age of those strata. But unravelling the dead past is not an easy task. One trying to unlock the “secrets of the cemetery of Nature’s dead,” walks on a shadowy road. His difficulties are many. It is like crossing a deep moat, climbing a steep wall.

Various methods have been applied to estimate the age of different periods of the earth’s history and much progress has been made toward a successful issue. Broadly, the procedure of different methods is the same. They do not differ in principle. “The rates of certain changes at the present day are determined as accurately as possible, and in imagination, the respective processes are traced backward in time until limiting conditions are arrived at.” Until the epoch-making discovery of radium, the two most outstanding methods used in calculating geologic time were (1) the rate of land erosion and deposition and (2) the rate of derivation of salt (sodium chloride) from the land and its accumulation in the oceans. Theoretically, it is simple to use the rate at which sediments are being deposited or solutions gathered into the ocean, as “geologic clocks” for estimating the length of past time. But in practice each method encounters its own difficulties and the results deduced give us at best only a rough idea of the immensity of time involved. I shall not deal with these individual estimates, but give

the mean of several, which is 100,000,000 years, speaking roundly. As stated before, it is only a rough estimate. Nevertheless, it confirms the fact that the earth is very old—indeed much older than is commonly believed. In 1650, Bishop Ussher, in his interpretation of the “In the beginning” of Genesis, estimated that the earth was created 4004 years before the birth of Christ. According to this view the earth is 5931 years old today. Many cosmogonists and even some geologists of the 19th century held this Biblical interpretation to be the age of the earth. Other ancient religions held that the earth was created much earlier than 4004 B.C. Hutton, one of the founders of Geology, in his studies, found “no vestige of a beginning—no prospect of an end.” One cannot help sympathizing with Hutton. Whoever has made a trip to the Grand Canyon of the Colorado River, Arizona, must remember the awe-inspiring depth of the Paleozoic strata and thousands of feet of Proterozoic sediments beneath them ([Pl. I](#)). If he has traveled farther north to the Cabinet Range, Montana, he must have carried with him an undying impression of the 35,000 ft. of the rocky monument, there built up by the Proterozoic seas.

LEAFLET 9.

PLATE II.



COMPARISON OF AN ANCIENT AND A MODERN MARINE ANIMAL OF ALLIED GROUPS.

- 1, A CAMBRIAN TRILOBITE (AFTER WALCOTT).
- 2, A MODERN HORSESHOE CRAB.

The present rate of denudation in the Hudson Bay region is one foot in about 47,000 years. How long then, has it taken the seas to lay down these miles of sediments, which are but a small fraction of the whole geologic column? Having considered all this, can we estimate, even approximately, the vast length of time that has elapsed between these periods of sedimentation, the so-called “gaps” or “breaks”—the torn, illegibly written pages of the history of the earth? Indeed, there are moments when all may feel that it is much beyond their comprehension. But, man, by nature, is at once humble and exalted. He is willing to admit his defeat, yet his thirst to conquer new knowledge, to know the truth, is never satiated.

The reason for the failure to arrive at an absolute result is not very far to seek. In computing geologic time, one has to calculate that which has elapsed, by some process in nature that takes place in one direction only and that does not change its rate when conditions alter. Whatever the method applied, be it the deposition of sediments, or the gathering of solutions, or the losing of heat by the sun and the earth, its rate of action should be uniform and uninterrupted. It should be independent of the changing conditions of the earth. Uniformity of the rate of action is the criterion for precise calculation. But we know that the past was quite different from the present. Different conditions have existed at different times during the earth’s history. The configuration of the earth, its climate, humidity, temperature and many other factors have varied from time to time and with them the rate of erosion, deposition and solution has either accelerated, diminished or ceased. The doctrine of uniformitarianism cannot be assumed in a changing world, even though our knowledge of the earth of the past can only be gained from a fuller study of the earth of the present.

When we consider the rate of sedimentation as a method for estimating geologic time, we take the total observed thickness of the geologic column, (estimated to be 70 miles) and divide it by the rate at which the sediments are now being laid down. But do we know this rate? By taking the average rate of sedimentation of nine large rivers^[1] now in existence and assuming it to be the rate at which sediments were deposited in the past, an approximate conclusion can be arrived at. But it will be noticed from the figures of the rate of sedimentation of different rivers that they are widely variable, the highest being many times greater than the lowest. Can we then use an average of nine figures so out of proportion and yet expect a reliable quantitative value? Furthermore, the rate of deposition along the coast, near the mouth of a large and active river, is much higher than on a coastline where no rivers empty their sediments into the sea. The rate is also largely controlled by the character of the sediments deposited. Sandstone and shale are more rapidly deposited than limestone. Moreover, just as there is no knowledge of the duration of time of erosion between periods of sedimentation, there is also no record of the amount of detritus that has fallen off the edge of the continental shelf during widespread emergence of the continents.

[1]

River	Drainage areas in square miles	Total tons annually	Ratio of sediment to water in weight	Height in feet of column of sedimen with a base of 1 square mile
Potomac	11,043	5,557,250	1: 3,575	4.0
Mississippi	1,244,000	406,250,000	1: 1,500	241.4
Rio Grande	30,000	3,830,000	1: 291	2.8
Uruguay	150,000	14,782,500	1:10,000	10.6
Rhone	34,800	36,000,000	1: 1,775	31.1
Po	27,100	67,000,000	1: 900	59.0
Danube	320,300	108,000,000	1: 2,800	93.2
Nile	1,100,000	54,000,000	1: 2,050	38.8
Irrawaddy	125,000	291,430,000	1: 1,610	209.0

Babb. Science, Vol. XXI, p. 343, 1893.

We have also no record of the vast quantity of shallow water sediments that were stirred up by the penetration of storm waves, and carried to abyssal depths by the currents and tides.

Similar uncertainties beset us when we consider the rate of chemical denudation, that is, the rate at which salts have been dissolved from the lands and accumulated in the oceans, as a measure of geological time. Here, again, we take the total amount of salts that is in the oceans today and divide it by the present rate of annual supply. We know with reasonable accuracy the quantity of salts in the oceans and if it were possible to assume the present rate of supply to be a true mean for all geological time, a satisfactory age of the oceans might be obtained. But it cannot be assumed as such. An assumption of this nature will only lead us from the domain of exactness to that of uncertainty. Aside from various other factors, neither the area of the continents, nor their relief was in the past the same as today. Consequently, the stream gradient and its power of dissolving salts from the land surfaces have not been the same. It is also not known how much salt the ocean derived from the shore line and from beds beneath the ocean, nor how much of the rock-salt beds on the earth that has been precipitated out of ocean water.

It is plain, therefore, that the rate of any process that is controlled by so many conditions cannot be used (even making generous allowances for irregularities and inaccessible data) as a reliable guide to evaluate geologic time. "It is a clock," says Harker, "which now hurries and now creeps or stands still, and it cannot be trusted as a timekeeper."

Any estimate based on the temperature of the earth, or of the sun, encounters similar practical difficulties, for the temperature of a body may not be constant. It may rise or it may fall. Further, the rate of change of temperature is controlled by a variety

of conditions, such as the amount of energy radiated, the supply of energy and so forth. Nor is there any record of the immense quantity of heat produced by igneous agencies and radio-activity.

Another estimate, one of the earliest, was based on the rate of life transformation in successive periods. The geological series were divided into twelve periods and it was believed that 20,000,000 years were required for an entire change in the species of each period, or 240,000,000 years in all. This does not include the time in which we have no record of plant or animal life.

There is no reasonable debate as to the passage of one species to another. It is clearly manifested in the succession of fauna found today all over the world in the sedimentary rocks. Even the most casual student of paleontology is convinced of this glaring truth. "The brutal cogency of a slab of fossils could be hated and fought, but could not be gainsaid." But when we are confronted with the question of setting a standard of measuring geologic time by means of this paleontological record, more precisely, through this biological process, we cannot help pondering over the grave uncertainty of the result. When we fix our gaze upon a trilobite, a three-lobed, crab-like creature ([Pl. II fig. 1](#)) that ruled the seas in the dim days of the Cambrian period ([p. 11](#)) and see that it was equipped with gills and swimming organs, with powers of digestion and excretion, with specific organs of circulation and reproduction and with motor and sensory nerves, and compare it with one of its tribe, a present day horseshoe crab, ([Pl. II fig. 2](#)) we do not find any noticeable progress in structure, in intricacy or in the degree of specialization. Yet the time that has elapsed since the Cambrian is, according to a moderate estimate, nearly 600,000,000 years! ([p. 11](#)). Geologic record testifies that evolution awaits environmental change, that animals in some way adjust themselves to their environment, either by discarding or modifying old characters or by acquiring new ones. Yet, what are known as "immortal" types, such as the brachiopods, *Lingula*, *Crania* and *Terebratula* ([Pl. III figs. 1-3](#)) or the pelecypods, such as *Pecten*, *Pinna* and *Arca* ([Pl. III figs. 4-6](#)) or the gastropods, such as *Pleurotomaria*, *Natica* and *Trochus* ([Pl. III figs. 7-9](#)), have withstood all possible environmental changes and have steadfastly held their own ever since we have records of their very early appearance on earth. On the contrary, we have records of types that have yielded so rapidly to change that their evolution is almost explosive. It is almost incomprehensible how, within such a limited period of time, fishes have changed into amphibians, amphibians into reptiles, and reptiles into birds and mammals ([Pl. IV figs. 1-4](#)). With these conflicting evidences staring us in the face, with the knowledge that the entire organic world has been subject to earth-wide periods of long stagnation and rapid intensive change, one may well ponder whether it is within our power to establish a standard for measuring geologic time on the evidence of life transformation. The study of the succession of faunas—the change of one species to another, can only indicate the magnitude of time involved. It cannot afford any basis, whatsoever, for a concrete expression of geologic time.

During the last three decades, a number of radio-active changes of one chemical element into another have been discovered and studies of certain minerals and rocks containing various radio-active elements have created means to calculate their age with remarkable accuracy. "A study of the various radio-active elements contained in minerals and rocks," says Harker, "has shown that it is possible, in certain favorable cases, to calculate directly their age in years."

The radio-active minerals are commonly found in igneous rocks. They are widely distributed all over the world. The parents of the whole series of radio-active elements are uranium and thorium. They possess the highest atomic weights of all known elements. Each of these parental elements transforms itself through a succession of changes. The final product of uranium is the formation of the metal lead and the gas helium. These transformations take place in one direction only, that is, from an element of higher atomic weight to an element of lower atomic weight. It has also been demonstrated beyond question that these transformations are unalterable by any process whatsoever and that they are independent of temperature, pressure or any other physical or chemical state. Temperatures up to 2,500 C. and pressures up to 600 tons per square inch have not been found to influence the rate of transformation. Time estimated on the basis of these processes, therefore, offers a more reliable result than that obtained by any other method hitherto known. Detailed descriptions of how the metal uranium slowly and regularly breaks down in a descending series into the metal lead and the gas helium, will be found in the literature on radio-activity. For our purpose, it suffices to say that according to Barrell, an atom of uranium which breaks up will ultimately give rise as a stable product to eight atoms of helium and one atom of lead. Since the rate of transformation is known, data for calculating the age of the mineral and with it the rock formation of which it is a part, can be obtained by measuring the quantity of helium and lead in the rock and comparing it with the quantity of uranium in the same volume of material. But, as helium is a gas, it is likely that a certain portion of it leaks out and consequently the estimate of age on the basis of how long helium had been in contact with uranium and lead is to be regarded as a minimum estimate. For example, the age of the mineral thorianite that occurs abundantly in the sands and gravels of Ceylon has been estimated to be 280,000,000 years, but the mineral is doubtless much older, as, ever since it was broken away from its original home in the pegmatite dikes of Ceylon, it lay exposed to the action of weathering and it was, therefore, very likely that during all these years a certain percentage of its helium contents had leaked away.

But estimates based on the lead ratios of radio-active minerals offer results consistent among themselves. That is, whenever fresh, primary, uranium-bearing minerals of the same geological age have been examined, the lead ratios are always found to remain constant. The value of the ratios increases or decreases as the geological age of the respective mineral increases or decreases. In other words, the lead ratios are in keeping with the geological age.

The procedure of applying the lead ratio in calculating geological time can be briefly stated thus: The rate of production of lead from uranium can be readily calculated. The rate at which helium is generated is accurately known and the quantity of lead liberated in the same time is approximately 6.5 times that of helium. In a year one gram of uranium produces 1.25×10^{-10} grams of lead, and at this rate 8,000^[2] million years will be required for the production of one gram of lead.

[2] A more recent and accurate computation reduces this 8,000 million years to 7,500 million years.

There is no serious difficulty in applying this method for measuring geologic time, except that it is necessary to determine whether the lead is of radio-active origin or original lead. The presence of original lead is likely to mar the constancy of the lead ratio essential for accurate results. But ordinary lead need not be confused with uranium lead, as the atomic weight of ordinary lead is 207.1 and that of uranium lead is 206.2. Values between these two figures imply a mixture of two types of lead. For reliable calculations, a series of fresh, primary minerals of the same geological age showing a constant lead ratio of atomic weight 206.2 needs to be examined.

The following table shows the geologic time that has elapsed between the first evidences of life and the present, as calculated by Barrell from radio-active data. The figures are his minimum and maximum estimates. It will be noticed from the figures in the table that the earliest life of which we have fossil records is about 1,500,000,000 years old. From this, it could be safely concluded that the inception of life on earth must have taken place much earlier. It is quite significant that each geological era, occasionally a geological period, has its characteristic grouping of life developed from the life of preceding periods. As we climb higher in the geological column, life becomes more and more complex and specialized. From the one-celled life of the Archeozoic it passes through the invertebrates—fishes—amphibians—reptiles—birds and mammals to man of the Recent time.

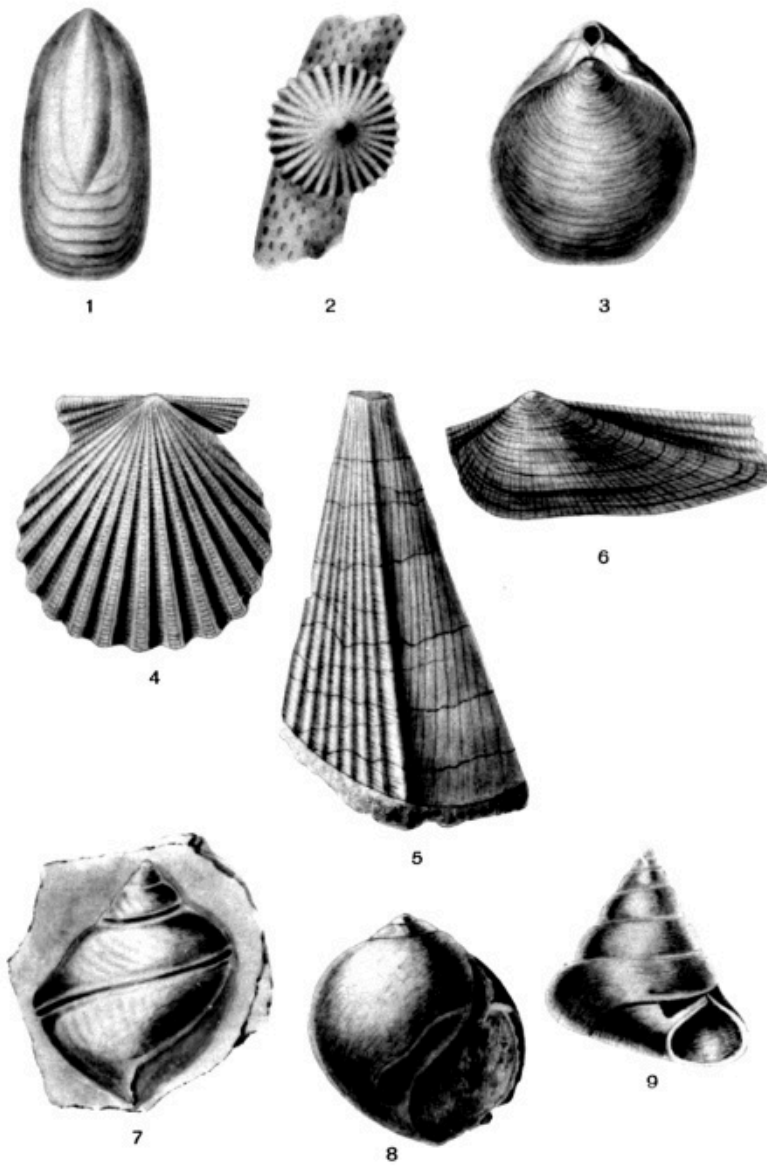
Since Barrell's publication of the estimates of geologic time as measured by means of radio-activity, some further studies have been made along the same line, but, as no generally accepted results have shown any marked differences from Barrell's results, it has been deemed advisable to use his age data as perhaps our present most adequate guide as to the length of geologic periods.

Although the measurable forces of radio-activity give on the whole a remarkably satisfactory time gauge and are doubtless more accurate than any method here discussed, it must not be considered that the ages given ([p. 11](#)) are absolute. The knowledge of geological time is of more importance for the comparative than for the absolute magnitude of the results obtained. Just as the study of astronomy gives us the conception of the vastness of space, so does the study of geology reveal to us that of the immensity of time.

SHARAT K. ROY.

LEAFLET 9.

PLATE III.



“IMMORTAL” TYPES.

1, LINGULA. 2, CRANIA. 3, TEREBRATULA. 4, PECTEN. 5, PINNA.
6, ARCA. 7, PLEUROTOMARIA. 8, NATICA. 9, TROCHUS.

(Figs. 1, 2, 7, after Hall, 3, 9, after Zittel, 4, 6, after Dall,
5, Pal. N. J. I, 8, after Cragin).

Drawings by Carl F. Gronemann.

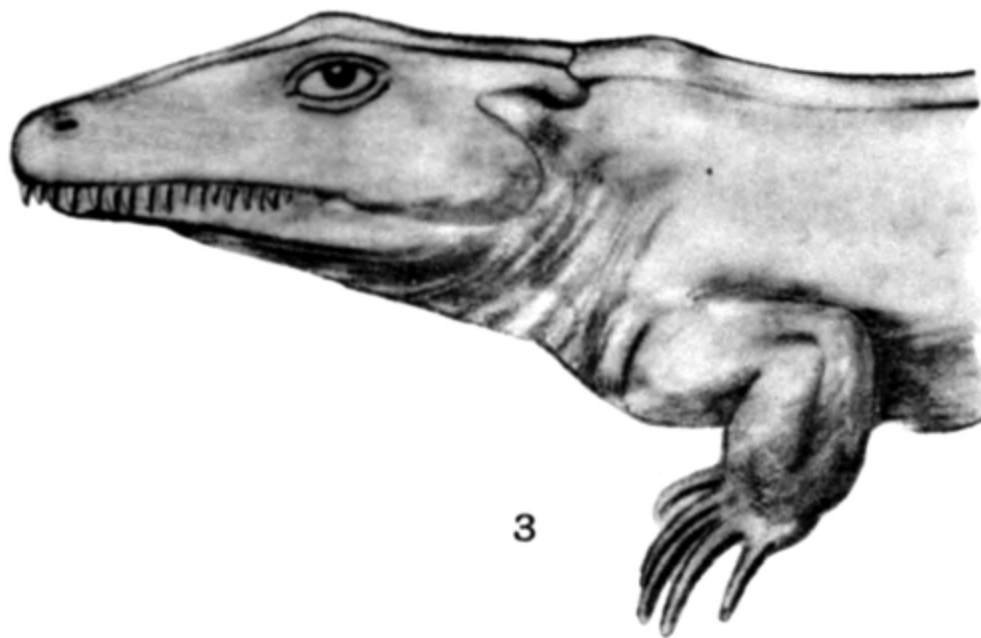
THE GEOLOGICAL TIME TABLE

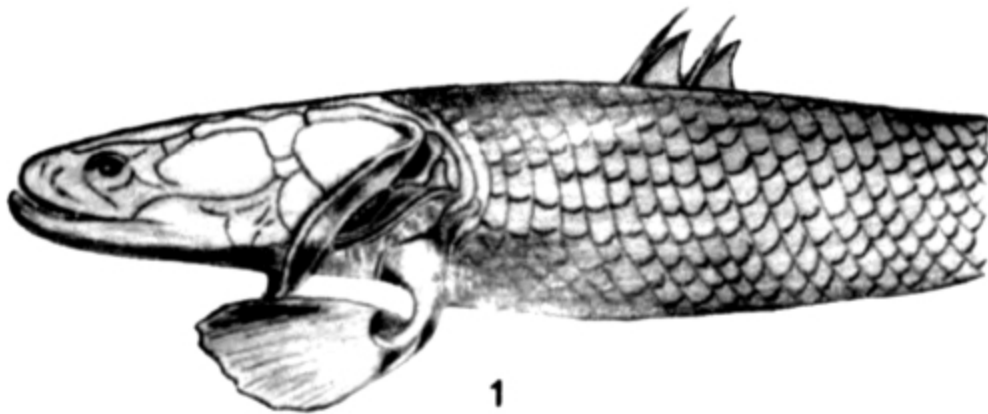
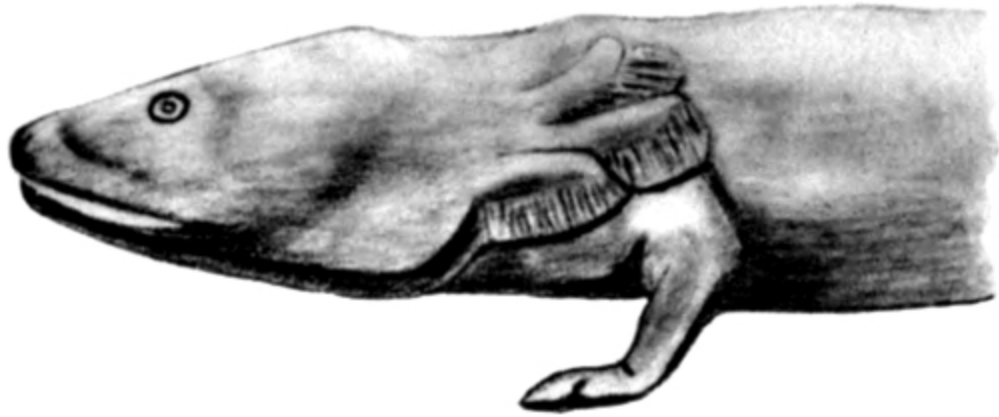
1.0 = One million years

ERAS	PERIODS	TIME SCALE (After Barrell)		CHARACTERISTIC LIFE
		Minimum	Maximum	
PSYCHOZOIC	Recent	1	1.5	AGE OF MAN
	Pleistocene			
CENOZOIC	Pliocene	7	9	AGE OF MAMMALS AND MODERN FLOWERING PLANTS
	Miocene	19	23	
	Oligocene	35	39	
	Eocene	55	65	
MESOZOIC	Cretaceous	95	115	AGE OF REPTILES
	Comanchian	120	150	
	Jurassic	155	195	
	Triassic	190	240	
PALEOZOIC	Permian	215	280	AGE OF AMPHIBIANS AND ANCIENT FLORAS
	Pennsylvanian	250	330	
	Mississippian	300	370	AGE OF FISHES
	Devonian	350	420	
	Silurian	390	460	AGE OF HIGHER SHELLED INVERTEBRATES
	Ordovician	480	590	
	Cambrian	550	700	
PROTEROZOIC	Systematic classification variable	925		AGE OF PRIMITIVE INVERTEBRATES
		LONG EROSIONAL INTERVAL		
ARCHEOZOIC		1500		DAWN OF UNICELLULAR LIFE, ALGAL FORMS REPORTED

BIBLIOGRAPHY

- BARRELL, J.—Rhythms and Measurements of Geological Time.
Bull. Geol. Soc. Am., Vol. 28, 1917, pp. 745-904.
- BECKER, GEO. E.—The Age of the Earth.
Smithsonian Misc. Coll., LVI., No. 6, 1910.
- CHAMBERLAIN, T. C.—Diastrophism and the Formative Processes,
XIII.
The Time over which the Ingathering of the Planetesimals was
Spread.
Jour. Geol., Vol. XXVIII, 1920, pp. 675-81.
- HARKER, A.—Geology in relation to the exact sciences, with an
excursus on geological time.
Nature, Vol. 95, 1915, pp. 105-109.
- HOLMES, A.—The Age of the Earth.
Harper and Bros., London and New York, 1913.
- JOLY, J.—Radioactivity and Geology.
Van Nostrand, N. Y., 1909;
- An Estimate of the Geological Age of the Earth.
Trans. Roy. Soc. Dublin, VII. 1899, pp. 23-66;
- The Age of the Earth. Phil. Mag., 6th ser.,
Vol. XXII, 1911, pp. 359-80.
- SOLLAS, W. J.—Presidential Address.
Quart. Journ. Geol. Soc., Vol. 65, 1909, pp. cxii;
Proc. Geol. Soc. of London, Sess. 1908-9, pp. i-cxxii.
- WALCOTT, C. D.—Geologic Time, as indicated by the sedimentary
rocks of North America.
Smithsonian Rep. 1893, pp. 301-334;
Jour. Geol. Vol. III, 1893, pp. 639-674.





EVOLUTION OF FISH TO MAMMAL-LIKE REPTILE.

1, FISH. 2, AMPHIBIAN. 3, REPTILE. 4, MAMMAL-LIKE REPTILE.

(Figs. 1, 2, after Klaatsch, 3, based on Williston, 4, based on Gregory).

Drawings by Sharat K. Roy.

PRINTED IN THE UNITED STATES OF AMERICA
BY FIELD MUSEUM PRESS

Transcriber's Notes:

The illustrations have been moved so that they do not break up paragraphs and so that they are next to the text they illustrate.

Typographical and punctuation errors have been silently corrected.

*** END OF THE PROJECT GUTENBERG EBOOK HOW OLD ARE
FOSSILS? ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.